MARC LABELLE

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Self-motivated and highly adaptable problem solver with a Master of Science in Physics and experience in multiple facets of software engineering, including web development, DevOps, military safety critical systems, mobile development, test automation, and RF systems engineering. Skilled at analyzing problems and finding solutions in a variety of languages. Proven ability to lead and work in a result-oriented team and meet demanding time goals. Strong written communication skills with experience developing technical proposals and documents. Named inventor on two U.S. patents.

PYTHON DJANGO FASTAPI GRAPHQL LABVIEW EMBEDDED C++ C# JAVA REACT SWIFT MYSQL POSTGRES AWS KAFKA DOCKER ELASTIC K8S

EXPERIENCE

SOFTWARE ENGINEERING CONSULTANT | LABELLE CONSULTING

Jun 2026 - Present

Consulting work across enterprise AI platforms and a biometric data-collection product.

- Built validation frameworks and integration tests for enterprise AI auth, connectivity, audit, and compliance workflows
- Improved platform security by identifying and resolving usage-enforcement vulnerabilities
- Built a biometric data platform with ESP32 hardware, MicroPython firmware, and a BLE-enabled React Native app
- Built Firebase infrastructure with Authentication, Firestore, Cloud Functions, and Storage for mobile-session delivery

STAFF SOFTWARE ENGINEER, R&D | ACV AUCTIONS

Mar 2023 - Present

Responsible for creating ground-up, business focused solutions to complex problems.

- Built hardware systems and a scalable, event-driven GraphQL and FastAPI backend powering ACV's AI inspection technology
- Developed scalable Elasticsearch service to efficiently match machine learning outputs with vehicle arbitration issues
- Architected configuration-as-code deployment of Tyk gateway with CI/CD testing to quickly and safely expose and track APIs
- Championed K8s implementation of SaltStack with an API wrapper and Nexus to securely manage embedded system OTA updates
- Architected an event-driven ML backend that used combinations of Kafka events to request GPU inference results via gRPC

LEAD SOFTWARE ENGINEER, R&D | ACV AUCTIONS

Sep 2021 - Mar 2023

Led a cross-functional team across backend, mobile, ML, and embedded systems; managed contractors and support.

- Lead and mentored team of backend, iOS engineers, contractors, and university students in all aspects of product development
- Designed and developed API microservices with Django, FastAPI, Postgres, MySQL, Docker, Kafka, Kubernetes, and Jenkins
- Developed iOS and embedded computer vision algorithms to automatically crop undercarriage images using OpenCV
- Developed embedded Linux audio and sensor logger; data used by ML models to reduce arbitration, $1M/yr savings (US Patent)
- Drafted software and test requirements documents for contractor deliverables

SENIOR ENGINEER | NSWC INDIAN HEAD, EOD DIVISION

Oct 2009 - Nov 2019

Lead developer responsible for several multiple programs, including web development, embedded analysis and physics modeling.

- Lead full-stack and devops developer, building applications for processing intelligence reports and analyzing data
- Developed and managed agile workflows, providing monthly reports and metrics to the sponsor using C3 for data visualizations
- Static and dynamic binary code analysis of safety-critical embedded devices to find bugs, vulnerabilities, and verifification
- Computational modeling of explosive effects for EOD and Counter-IED methods and devices (US Patent)
- Lead designer and developer of automated test systems for RF signals, communications, system timings, and electrical properties
- Identified and developed custom Counter-IED solutions for EOD technicians using PIC, Atmel, Arduino, Embedded C and Assembly
- Worked with contracts and external vendors to expand and develop concepts, designs, testing and acceptance reviews
- Developed embedded data acquisition logger software that recorded various real-time test data streams simultaneously

EDUCATION

MASTER OF SCIENCE (ASTROPHYSICS CONCENTRATION) | NEW MEXICO INSTITUTE OF MINING AND TECHNOLOGY — May 2009

BACHELOR OF PHYSICS (EXPERIMENTAL CONCENTRATION) | STATE UNIVERSITY OF NEW YORK AT FREDONIA — May 2007